



UNIVERSITY OF
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COMP575

Tutorial 4

Radial Basis Functions and Radial Basis Function Networks

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Today's Content

- Overview of
 - RBF features
 - RBF interpolation
 - RBF neural networks

Learning Outcomes

1. Define an RBF and explain radial and local receptive field behaviour in terms of distance to a centre (denoted as c)
2. Formulate an RBF network/RBF regression model as a linear model in a nonlinear feature space
3. Construct the RBF matrix (denoted as Φ) for a dataset
4. Fit the output weights w using least squares (aka pseudoinverse)
5. Describe and apply regularised least squares for generalisation
6. Select centres and widths using two strategies (random subset, k-means, heuristic, global σ , etc)
7. Implement and evaluate a simple 1D RBF



Motivation and Definitions

- Often, datasets aren't linearly separable in an input space $\mathbb{R}^d$
 - The input vector would be expressed as $x \in \mathbb{R}^d$
- Therefore, we map inputs into features $\phi(x) \in \mathbb{R}^b$, then we can apply a linear classifier or a regressor
 - $\phi(x) \in \mathbb{R}^b$ is the vector of transformed features
- The linear decision function in ϕ -space is defined as:
$$g(x) = w^T \phi(x)$$
- RBF networks use weights
 - Coefficients that control how much each radial basis function contributes to the final output
 - Intuitively, the weight vector of b is expressed as $w \in \mathbb{R}^b$

Pattern Separability and ϕ -separability

- A binary dataset is ϕ -separable if $\exists w$ such that

$$\begin{aligned}w^\top \phi(x) &\geq 0 \forall x \in C_1, \\w^\top \phi(x) &< 0 \forall x \in C_2\end{aligned}$$

- In ϕ -space, the hyperplane corresponds to the boundaries in input space
 - Curved boundaries in input space. In ϕ -space it's a hyperplane (which is linear) which when mapped back to input space, it produces a nonlinear decision boundary
 - Some separation examples:
 - Linear (plane)
 - Spherical
 - Quadratic
- In our binary dataset example, the hyperplane $w^\top \phi(x) = 0$ splits the ϕ -space into two halves
 - Aim: Can we estimate the separating boundary using RBF?

The XOR Problem

- Becomes linearly separable with RBF
 - Not possible in 2D
 - We can use pattern separability to map the patterns so that they become linearly separable
 - We don't need to increase the dimensionality

- Process

- Choose two RBF features centred at (1,1) and (0, 0)

$$\varphi_1(x) = \exp(-\|x - c_1\|_2^2), \quad c_1 = (1,1)^T$$

$$\varphi_2(x) = \exp(-\|x - c_2\|_2^2), \quad c_2 = (0,0)^T$$

- Generate new truth table
 - Hint: we often use a Gaussian radial shape. The above functions are similar (see lecture notes)
 - Note: the decision boundary will always be $w^T \varphi(x) + w_0 = 0$
 - The weighted sum with positive and negative weights can still cross 0, so no ambiguous predictions

Definition of RBF

- Radial symmetry around the centre c
 - In other words... RBF is a “bump” centred at point c
- Activation depends only on distance from x to c
 - In other words... the output depends only on how far x is from c

- Mathematical definition:

$$\phi(x) = \varphi(\|x - c\|_2)$$

- RBFs create local receptive fields
 - These are basically neighbourhoods around centres
 - They detect points falling into their “neighbourhoods”

Interpolation

- Interpolation definition: the process of constructing a function that matches the given data points **exactly**

- Example:

- Data points (x_i, y_i)
- Find function $F(\cdot)$ such that

$$F(x_i) = y_i, \quad i = 1, \dots, N$$

- Mathematical definition of the exact fit:

$$F(x) = \sum_{i=1}^N w_i \varphi(\|x - c_i\|_2)$$

- Note: exact interpolation will likely overfit noisy data

Interpolation

- We then setup the interpolation matrix (denoted as ϕ)
- Enforce $F(x_j) = y_j$ (we want each training point to pass through the function)
 - Define interpolation matrix $\phi_{ji} = \varphi(\|x_j - x_i\|_2)$
 - Evaluated for each training point
- Interpolation constraints: $\phi w = y$ (exact matching)
- Note (follow-up from previous slide's note): if ϕ is invertible, then $w = \phi^{-1}y$ and we get a unique interpolating function
 - Since exact interpolation can overfit, we often use $M \ll N$ bases and least squares or regularisation instead of exact interpolation
 - ϕ is invertible when it's a square matrix and there exists a matrix ϕ^{-1} such that $\phi^{-1}\phi = I$ and $\phi\phi^{-1} = I$



From Interpolation to RBF Neural Networks

- Interpolation uses $M = N$ centres (expensive, often redundant)
 - M number of basis functions
 - N data points
- RBF NN uses $M \leq N$ basis functions
 - Typically $M \ll N$

$$F(x) = w_0 + \sum_{i=1}^M w_i \phi_i(x)$$

- Gaussian radial shape (most widely used):

$$\phi_i(x) = \exp\left(\frac{-\|x - c_i\|_2^2}{2\sigma_i^2}\right)$$

- Why?
 - Fewer bases means faster execution and less redundancy
 - Approximation works better on noisy data

Model Matrix for Training

- What is a model matrix?
 - AKA design matrix, collects the values of the model's features for every training example
 - It's the matrix of your explanatory variables
- Take a training set $\{(x_n, y_n)\}_{n=1}^N$
- Define the model matrix as $\phi_{ni} = \phi_i(x_n) = \varphi(\|x_n - c_i\|_2)$
- Represent predictions as $\hat{y} = \phi w$
- Training $\hat{w} = \phi^+ y$



Regularisation

- Definition: adding an extra term to the training objective that discourages overly complex solutions
- If we only minimise the error (like in the previous example)...
 - We end up overfitting
 - We will have large weights
 - Etc (beyond the scope of today's tutorial)

Centres and Widths

- Centres (c_i): fixed points. Picked either:
 - At random
 - Using a clustering algorithm
 - One for each data point
- Widths (σ_i): controls how far the influence spreads
 - If σ too small, we can overfit on the data
 - If σ too large, we can underfit

Learning Strategies: Random Subset

- Centre selection: Randomly pick training samples as centres c_i
- Width:
 - Compute d_{max} (the max distance between chosen centres)
 - Choose a global width based on spacing (goal is to avoid too peaky/flat surface)
- Using the resulting max distance, set $\sigma = \frac{d_{max}}{\sqrt{2M}}$
- Weights: calculated using least squares fit using the pseudoinverse

Learning Strategies: Self-Organised

- Centre selection: use clustering-based technique (e.g. k-means)
 - As a result, there will be less bases and therefore the RBF network will be quicker to execute
- The centroids of these clusters will serve as c_i for each cluster
- Width: either per-centre σ_i from within-cluster variance OR avg distance to neighbours
- Weights: calculated using least squares (same as before)

Learning Strategies: Supervised Optimisation

- Optimises all 3 params (c_i, σ_i, w_i) by minimising the error function

$$E(\{w_i, c_i, \sigma_i\}) = \frac{1}{2} \sum_{j=1}^N e_j^2$$
$$e_j = \sum_{i=1}^M w_i \exp\left(\frac{-\|x_j - c_i\|_2^2}{2\sigma_i^2}\right) - y_j$$

- We use stochastic gradients with three different learning rates $\eta_c, \eta_\sigma, \eta_w$ to learn the centre, the width and the weight

Demo Time!



Summary

- RBFs are distance-based nonlinear basis functions
- Interpolation gives exact fit
 - Expensive, overfits
- Networks give approximations
 - Cheap, flexible
- Easy to train
- Centres and widths require some selection strategy

Glossary

... JUST AN FYI!

Glossary – Sets, Spaces, Dimensions

- $\mathbb{R}^d$: d -dimensional real input space (e.g. the original feature space)
 - d : number of input variables
- $\mathbb{R}^b$: b -dimensional transformed feature space (e.g. the mapped feature space)
 - b : number of features (input variables post mapping)

Glossary – Data and Indexing

- (x_i, y_i) : the i th data point (input x_i , target y_i)
- x_n : the n th input in the training set
- y_n : the n th target
- N : number of training samples (or data points)
- M : number of RBF bases (or hidden units in networks)

Glossary – Feature maps & Basis Functions

- $\phi(x)$: feature vector (the mapping of input x into the feature space)
- $\phi_i(x)$: the i th feature or RBF unit activation
- $\varphi(\cdot)$: the radial shape function (e.g. Gaussian)
 - Takes a distance, returns an activation
 - Example: $\phi(x) = \varphi(\|x - c_i\|_2)$

Glossary – Distances and lin alg operators

- $\|v\|_2$: L2 norm of vector v
- $\|x - c\|_2^2$: squared Euclidean distance between input x and centre c
- $(\cdot)^T$: transpose of a matrix
- $\exists$: “there exists”
- $\forall$: “for all”
- $\in$: “element of”

Glossary – Centres, Widths

- c_i : centre of the i th RBF unit
- σ_i : the width (or spread) of the i th RBF unit
- d_{max} : max distance between chosen centres
 - Used in heuristics for σ
- (EXTRA) Gaussian RBF: $\exp\left(\frac{-\|x_j - c_i\|_2^2}{2\sigma_i^2}\right)$

Glossary – Models, Predictions, Weights

- $F(x)$: model function
- $g(x)$: linear decision function
- w : vector of output weights
 - w_i : weights for basis function i
 - w_0 : bias term
- $\hat{y}$: vector of predicted outputs
- $\hat{w}$: estimated weight

Glossary – Matrices

- ϕ : design matrix
- ϕ^+ : pseudoinverse
- I : identity matrix

Glossary – Error, Optimisation, Regularisation

- e_j : residual (error on training example j)
 - More formally $e_j = \hat{y}_j - y_j$
- E : total squared error objective
 - On slide 16: $E = \frac{1}{2} \sum_{j=1}^N e_j^2$
- λ : regularisation strength (not used in these slides – please see lecture notes)
- η : learning rate

Q&A

